

Introduction

CPU / Processor

Transistors
0v(0) and 5v(1)

0's and 1's

1940's ==> write the code for addition

Add ==> 01010100110

Sub ==> 01011110101

Program / Code ==> written using 0's and 1's ==> Machine Level Language

Add ==> 01010100110

Sub ==> 01011110101

Problem ==> MLL was difficult to remember

1950's ==> solution

Add ==> ~~01010100110~~

Sub ==> ~~01011110101~~

add 10 20

sub 30 10

Assembler

MLL

CPU

Assembly Level Language

1960's

Addition ==> 01010101010 ==> Add ==> +

Subtraction ==> 01010101010 ==> Sub ==> -

High Level Language ==> written using symbols and english words

1960's till 2026 ==> HLL

C ==> English words + symbols ==> HLL

C++ ==> English words + symbols ==> HLL

Java ==> English words + symbols ==> HLL

HLL ==> 10 + 20(HLL) ==> COMPILER ==> MLL ==> CPU

1960's ==> HLL was introduced

1962 ==> BCPL ==> Martin Richards (Basic Combined Programming Language)

--> unstructured language(we cannot reuse the code)

--> more memory / space

Write a calculator program

addition

subtraction

Multiplication

Division

addition

Multiplication

1969 ==> Ken Thompson ==> B

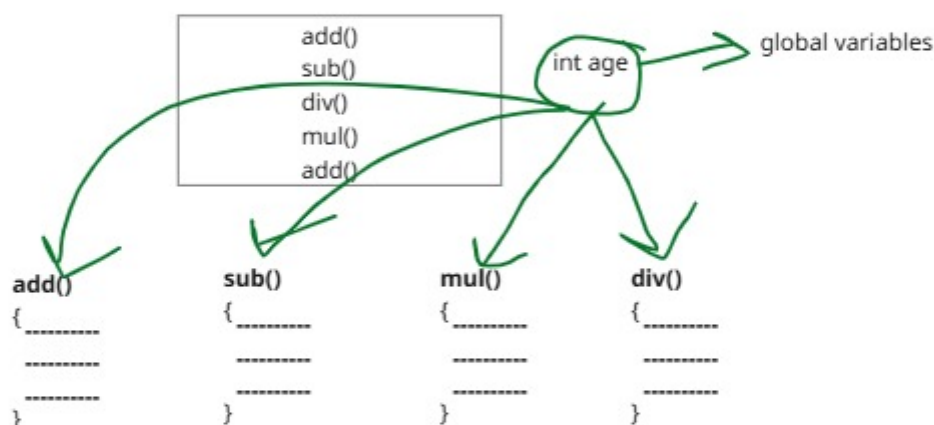
--> Less Memory

--> Unstructured

1972 ==> Dennis Ritchie ==> C

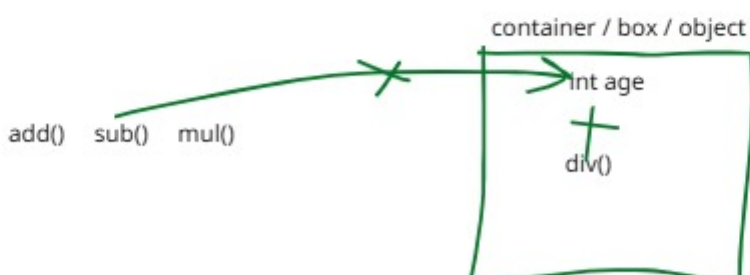
--> Structured

--> Less memory



1982 ==> Bjarne Stroustrup ==> C++

--> invented a new language that solves the problem of C language



problem of C++ ==> non portable

JAVA ==> unofficially introduced ==> 1992 ==> officially introduced ==> 1996

To solve this problem ==> JAVA

==> supports objects

==> less memory

==> portable